

CAPE TOWN · ENERGY · DATA · AI

Tiffany Woodley

Engineer, data scientist and technology leader across energy, machine learning and AI

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Professional Profile

Engineer, data scientist and technology leader — from battery and solar modelling under load shedding to production machine learning and AI systems.

I started in renewables software, building SolaFuture's first battery sizing and financial modelling tools and a peak-shaving product sold commercially. At Solink, through South Africa's load-shedding years, I owned the company's battery financial modelling strategy — Eskom-driven scenarios, sizing, solar forecasting with machine learning, control algorithms and cost optimisation — leading the approach, mentoring engineers and reviewing every proposal. I later led systems development there, shipping monitoring and consulting platforms end to end.

That path sits on dual master's training in data science and advanced analytics, including reinforcement learning for peak shaving, and applied ML engineering at Bytefuse. I now lead data science at Sand Technologies: production ML and MLOps, forecasting frameworks, scenario-based decision making and generative AI workflows. I turn quantitative insight into software and decisions teams can trust.

Energy and modelling

Battery and Solar Financial Modeling
Battery System Control Systems
Forecasting
Optimisation

Data science and AI

Machine learning
Generative AI
LLM workflows
MLOps

Technology

Software engineering
Data engineering
Solutions Architecture
Technical Leadership

Timeline

2025 – Now
Full time

Senior Data Scientist · Sand Technologies

Healthcare – Lead Data Scientist and Engineer

- Lead data science and engineering for the Healthcare product — pipelines, data architecture and analytical delivery.
- Support scenario-based decision making — structuring options, trade-offs and outcomes so teams can choose under uncertainty.
- Run projects end to end: frame the problem, choose the approach, design the system and ship with cross-functional teams.
- Translate messy operational requirements into practical, production-ready data solutions.

Central R&D – Data Science and AI

- Work across forecasting model approaches and focus on the production framework — taking models into reliable systems with MLOps, monitoring and operational handover.
- Build reusable data-quality and drift monitoring for MLOps — catch distribution shifts before models degrade in production.
- Design monitoring architecture that helps organisations keep production ML reliable over time.
- Bring generative AI and LLMs into analytical and software workflows to speed up delivery of robust applications.

Software engineering | Machine learning | Data engineering | Forecasting | Optimisation | Generative AI | LLM workflows | LLMs | MLOps | Technical Leadership | Solutions Architecture | Data quality | Model monitoring

2024 – 2025
Full time

Head of Systems Development · Solink

- Architected Soltrak, a high-availability energy monitoring platform designed for fault-tolerant operations.
- Built internal platforms that automated operations — reporting frameworks and energy financial models embedded in day-to-day processes.
- Owned new product delivery end to end, including an automated energy consulting platform from architecture through build.

Software engineering | Battery and Solar Financial Modeling | Solutions Architecture | Technical Leadership | Node.js | Postgres | Appscript | AWS

2023 – 2024
Full time

Systems Engineer · Solink

- Built the battery financial modelling strategy the company used for proposals — simulating Eskom se Push load-shedding schedules, seasonality and multi-year scenario paths, then switching to arbitrage or grid-avoidance dispatch when shedding eased so projects showed full financial sensitivity.
- Led the modelling approach across the team, mentored engineers on it, and reviewed every proposal that came through.
- Optimised system sizing with the financial models, showing where oversizing for rare events hit diminishing returns and where the practical cut-off sat.
- Optimised solar forecast operating points to reduce costs, linking prediction outputs to dispatch and commercial outcomes.
- Wrote the control algorithms behind the energy models and delivered a client peak-shaving optimisation solution.
- Built machine learning models for solar production forecasting and advanced learned prediction to support system performance, financial modelling and consulting workflows.
- Automated operations reporting and added anomaly detection so underperformance surfaces early from distributional shifts.
- Defined the end-to-end architecture for an automated energy consulting platform.
- Tightened delivery cycles by introducing agile ways of working across the engineering team.

Software engineering | Battery and Solar Financial Modeling | Battery System Control Systems | Forecasting | Machine learning | Optimisation | Technical Leadership | Solutions Architecture | Node.js | Postgres | Appscript

2022 – 2023
Full time

Machine Learning Engineer · Bytefuse.ai

- Built reinforcement-learning controllers for traffic signals to improve flow and reduce congestion.
- Delivered machine learning solutions for energy-industry problems.

Software engineering | Machine learning | Optimisation | Reinforcement learning

2020 – 2022
Education

MSc Advanced Analytics · University of Cape Town

- Thesis: reinforcement learning for peak shaving — applying RL to a real energy optimisation problem.
- Advanced statistics and modelling: multivariate statistics, regression, simulation, optimisation and machine learning.

Machine learning | Optimisation | Battery System Control Systems | Statistics | Reinforcement learning

2019 – 2020
Education

MSc Data Science · University of Cape Town

- Core training in supervised and unsupervised learning, industry data science, databases and visualisation.

Machine learning | Data engineering | Data visualisation | Databases

2017 – 2019
First roles

Junior Software Engineer · SolaFuture Energy

Intern in 2017, then Junior Software Engineer 2018–2019.

- Created SolaFuture's first battery sizing software — a Python-backed financial model that simulated projects, ran scenarios and generated all client battery proposals.
- The tool became the company's standard proposal engine and continued in use for years after I left.
- Built a peak-shaving model that was sold as a commercial product, including the forecasting needed to drive peak-shaving algorithms.
- Built early renewables software on Python, SQL, VBA and Linux systems.

Software engineering | Battery and Solar Financial Modeling | Battery System Control Systems | Forecasting | VBA | Linux server | SQL

2012 – 2018
Education

BEng Electrical & Electronic Engineering · Stellenbosch University

Electrical and electronic engineering foundation — the base for later work in energy systems and data science.

Engineering | Systems | Solutions Architecture

Open to conversations about energy systems, applied AI and building things that hold up in production. · tiff.woodley@gmail.com